

Chapter 13 :

Feedback in Amplifiers

13.1 Introduction

In Chapter 4, we have already found the term **feedback** in context of biasing. We described there that the emitter resistor (R_E) provides stability in a mechanism called negative feedback. In this chapter, we will learn feedback elaborately. Feedback theory is commonly used in amplifiers, oscillators and some other complex systems in different technical fields. So it is important to have clear concept of feedback.

Feedback is the process by which a fraction of output signals either voltage or current, is used as an input or combined with the input. If this fed back fraction is opposite in value or phase (or anti-phase) to the input signal, then the feedback is said to be **Negative Feedback**, or **degenerative feedback**. That is, if the feedback signal weakens the original input signal, then it is called negative feedback. On the other hand, if the fed back fraction adds up with the input or in phase with the input signal, then the feedback is said to be **Positive Feedback**, or **regenerative feedback**. That is, if the feedback signal strengthens the original input signal, then it is called positive feedback. Negative feedback is mainly used in amplifiers, while positive feedback is used in oscillators, multivibrators, comparators etc.

13.2 Feedback theory

The general structure of a feedback amplifier may be depicted by the signal flow diagram as in Fig.13-1. Here, x may be a voltage or current signal.

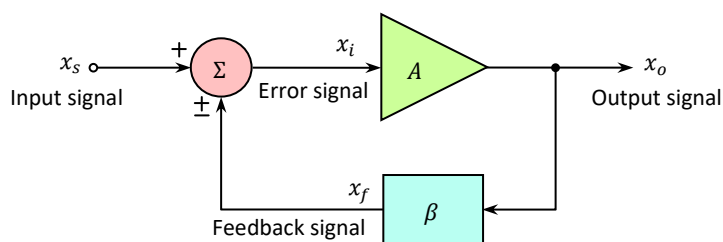


Fig.13-1: General feedback circuit

The basic amplifier (to which the feedback is applied) is supposed to be a unilateral amplifier. The gain of the amplifier without feedback is A and this gain is called **open-loop gain**. This amplifier amplifies the input signal (x_i), i.e. $x_o = Ax_i$.

The feedback unit (β) may vary from a simple direct connection to a complex circuit. This block is connected between input and output of the amplifier and produces a feedback path. When feedback is applied to a system (or any amplifier), it is called **closed-loop system**. The feedback unit produces a feedback signal x_f (either voltage or current) which is proportional to output signal x_o such that $x_f = \beta x_o$. The fraction $\beta = x_f/x_o$ is called **feedback factor**. The feedback signal (x_f) has been applied to a summing unit, that combines the feedback signal, x_f , with the input signal, x_s . Depending on the polarity of x_f , the summing unit may add or subtract x_f from the input signal x_s . Thus the summing unit produces a resultant signal $x_i = x_s \pm x_f$ which is called the **error signal** (generally in control engineering). This error signal (x_i) is applied to the amplifier. So the output of the amplifier will be,

$$x_o = Ax_i = A(x_s \pm x_f)$$

But, we know $x_f = \beta x_o$. Putting this value into the above equation we get,

$$x_o = A(x_s \pm \beta x_o)$$

$$\text{or, } x_o = Ax_s \pm A\beta x_o$$

$$\text{or, } x_o \mp A\beta x_o = Ax_s$$

$$\text{or, } x_o(1 \mp A\beta) = Ax_s$$

Therefore, the gain of the amplifier with feedback will be,

$$A_f = \frac{x_o}{x_s} = \frac{A}{1 \mp A\beta} \quad (13-1)$$

$$\text{or, } A_f = \frac{A}{1 + A\beta} \text{ (negative feedback)} \quad (13-2)$$

$$\text{or, } A_f = \frac{A}{1 - A\beta} \text{ (positive feedback)} \quad (13-3)$$

Here, A_f is the gain of the amplifier with feedback. This gain is also called **closed-loop gain**. The product of amplifier's gain (A) and feedback factor (β) is called **loop-gain**. That is the loop-gain of this amplifier is $A\beta$.

Depending on the polarity (or phase) of x_f , there are two types of feedback. If x_f is positive, it is called positive feedback, and if x_f is negative it is called negative feedback.

Example 13-1

An amplifier has a voltage gain of 100. If negative feedback is applied the overall gain falls to 10. (i) Calculate the feedback factor (β). (ii) If the value of β is maintained, calculate the required gain of the amplifier to make the overall gain 11.

Solution:

(i) In the first case, $A = 100$, and $A_f = 10$.

Using Equ.(13-1) $A_f = \frac{A}{1 + A\beta}$ or, $10 = \frac{100}{1 + 100\beta}$

or, $10 + 1000\beta = 100$

or, $\beta = \frac{90}{1000} = 0.09$ [Ans.]

(ii) For the second case, $\beta = 0.09$, and $A_f = 11$.

Using Equ.(13-1) $A_f = \frac{A}{1 + A\beta}$ or, $11 = \frac{A}{1 + A \times 0.09}$

or, $11 + 0.99A = A$

or, $0.01A = 11$

or, $A = \frac{11}{0.01} = 1100$ [Ans.]

Comments: Here a large negative feedback is used. So open loop gain 100 falls to only 10. For this large negative feedback A_f has been very stable. So for a small change in A_f (1) a large change in A (1000) is required.

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### 13.3 Types of Feedback Amplifiers

Already we have mentioned that depending on the phase (or polarity) of feedback signal, there are two types of feedback. These are: Negative Feedback, and Positive Feedback.

#### 13.3.1 Positive Feedback Amplifier

If the feedback signal is mixed with the input signal in such a way that the overall input to the amplifier increases, then it is called *positive feedback*. As the input signal is increased, positive feedback increases the gain of the amplifier as shown in Fig.13-2(a). Positive feedback is also

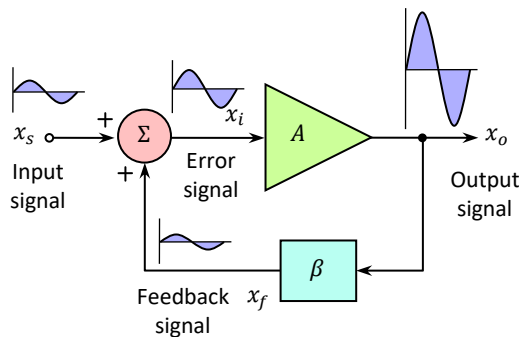


Fig.13-2 (a): Positive feedback

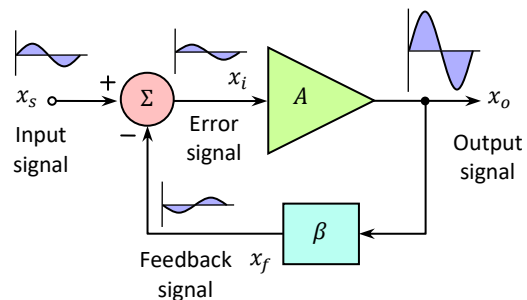


Fig.13-2 (b): Negative feedback

called **direct feedback**. As positive feedback generates distortion in the output signal, it is not often used in amplifiers. But, it amplifies power of the original signal and can be used in oscillator circuits. The main disadvantages of positive feedback are given below:

- Lower input impedance
- Unstabilized voltage gain
- Increased output impedance
- Increased noise, and
- Increased nonlinear distortion

### 13.3.2 Negative Feedback Amplifier

If the feedback signal is mixed with the input signal in such a way that the overall input to the amplifier decreases, then it is called **negative feedback**. As the input signal is decreased, negative feedback decreases the gain of the amplifier as shown in Fig.13-2(b). Negative feedback is also called **inverse feedback**. Because negative feedback provides many advantages, it is regularly used in amplifiers. The main advantages of negative feedback are given below:

- Better stabilized voltage gain
- Higher input impedance
- Improved frequency response
- Lower output impedance
- Reduced noise
- More linear operation

## 13.4 Increased Stability

The gain of an amplifier may change due to temperature variation or change in device parameters when they are replaced. Negative feedback provides stabilization of the gain. From Equ.(13-2), the gain of an amplifier with negative feedback will be,

$$A_f = \frac{A}{1 + A\beta}$$

To get stabilized gain, we have to design the amplifier with a large value of  $A\beta$ . Then, neglecting 1 in the denominator, we can write the above equation as,

$$A_f \approx \frac{A}{A\beta} \quad \text{or,} \quad \boxed{A_f = \frac{1}{\beta}} \quad (13-4)$$

Thus, the closed loop gain of the amplifier depends only on the feedback factor ( $\beta$ ), that is, the characteristics of the feedback network not on the gain of the amplifier ( $A$ ). Generally the feedback network is designed using resistors (a voltage divider network). Resistors can be selected very precisely with almost zero temperature coefficients. Thus, though the open-loop gain of the amplifier will vary due to temperature change, device parameters change, and frequency change, the closed loop-gain will not change as it depends only on  $\beta$ .

### Example 13-2

Negative feedback is applied to an amplifier having open loop gain of 500. The value of feedback factor is 0.1. If the open loop gain of the amplifier decreases by 20%, due to temperature variation, determine the percentage of variation in the closed loop gain.

**Solution:**

Given:  $A = 500$ ,  $\beta = 0.1$ . Using Equ.(13-2), the gain of the amplifier with feedback will be

$$A_f = \frac{A}{1 + A\beta} = \frac{500}{1 + 500 \times 0.1} = 9.80$$

Due to the temperature change, the new gain will be  $A = 500 \times (100 - 20)\% = 400$ . So the new closed loop gain will be,

$$A'_f = \frac{A}{1 + A\beta} = \frac{400}{1 + 400 \times 0.1} = 9.756$$

$$\% \text{Variation} = \frac{9.80 - 9.756}{9.80} \times 100 = 0.45\% \text{ [Ans.]}$$

**Comments:** Due to negative feedback, the overall gain of the amplifier has decreased to 9.80 from 500 (disadvantage). With 20% change in open loop gain, the overall gain has changed by only 0.45%.

## 13.5 Reduction in Non-Linear Distortion

Generally, a large signal amplifier suffers from non-linear distortion. As the operating point swings over a large portion of the load-line, the amplifier provides different voltage gain at different points of the signal's cycles. The use of negative feedback in amplifiers reduces the non-linear distortion.

Let,

$v_D$  = distortion voltage generated by the amplifier without feedback

$v'_D$  = distortion voltage generated by the amplifier with feedback

$A$  = open loop gain of the amplifier

$\beta$  = feedback factor

$$\text{Suppose,} \quad v'_D = x v_D \quad (13-5)$$

Now, the portion of  $v'_D$  that will be fed back into the amplifier is  $\beta v'_D = \beta x v_D$ . This voltage will be amplified by the amplifier. So, the amplified distortion voltage will be  $A\beta x v_D$ . Due to the negative feedback this voltage will be subtracted from  $v_D$ . Thus, the new distortion voltage will be,

$$v'_D = v_D - A\beta x v_D \quad (13-6)$$

From Equ.(13-4) and Equ.(13-5) we get,

$$x v_D = v_D - A\beta x v_D$$

or,

$$x v_D (1 + A\beta) = v_D$$

or,

$$x = \frac{1}{1 + A\beta}$$

Putting this value of  $x$  into Equ.(13-4),

$$v'_D = \frac{v_D}{1 + A\beta}$$

Thus, we find that the negative feedback reduces the distortion voltage by a factor  $(1 + A\beta)$ . The situation is illustrated in Fig.13-3, for an amplifier with  $180^\circ$  phase shift. Fig.13-3(a) shows the input and output waveforms when there is no feedback deployed in the amplifier. Due to the non-linearity of the amplifier the negative half cycle of the output has been distorted. Fig.13-3(b) shows the same input and output waveforms when feedback is deployed in the amplifier. Now the gain of the amplifier has been decreased, the distortion has also been reduced.

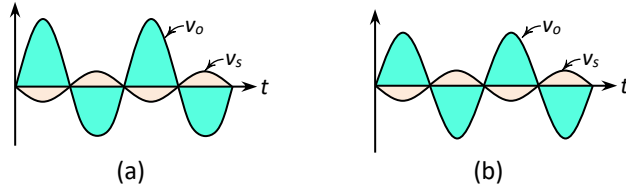


Fig.13-3: Input & output waveforms of an amplifier: (a) With no feedback, large distortion, (b) with feedback less distortion

### 13.6 Effect of Negative Feedback on Lower Cut-off Frequencies

The frequency ( $f_L$ ), in the lower frequency range, at which the voltage gain falls to  $1/\sqrt{2}$  of its mid-frequency range is called the lower cut-off frequency. The gain of an amplifier without feedback can be represented in terms of this lower cut-off frequency as [Equ.(10-)],

$$A_L = \frac{A_m}{1 - j(f_L/f)} \quad (13-7)$$

Where,  $A_m$  is the mid-frequency voltage gain and  $f_L$  is the lower cut-off frequency.

Now, if negative feedback is used in the amplifier with feedback factor  $\beta$ , the mid-frequency gain will be,

$$A'_m = \frac{A_m}{1 + \beta A_m} \quad (13-8)$$

The voltage gain in terms of lower cut-off frequency can be represented as,

$$A'_L = \frac{A_L}{1 + \beta A_L}$$

Putting the value of  $A_L$  from Equ.(13-6) we get,

$$A'_L = \frac{A_m/[1 - jf_L/f]}{1 + \beta A_m/[1 - jf_L/f]}$$

or,

$$A'_L = \frac{A_m}{1 - jf_L/f + \beta A_m}$$

or,

$$A'_L = \frac{A_m}{1 + \beta A_m} \times \frac{1}{1 - j f_L / f (1 + \beta A_m)}$$

Using Equ.(13-7),

$$A'_L = \frac{A'_m}{1 - j f'_L / f}$$

where,

$$\boxed{f'_L = \frac{f_L}{1 + \beta A_m}} \quad (13-9)$$

Equ.(13-8) shows that due to the negative feedback the lower cut-off frequency has been changed to  $f'_L$ . Here the lower cut-off frequency has been decreased, due to the negative feedback, by a factor  $(1 + \beta A_m)$ .

### 13.7 Effect of Negative Feedback on Higher Cut-off Frequency

The frequency ( $f_H$ ), in the higher frequency range, at which the voltage gain falls to  $1/\sqrt{2}$  of its mid-frequency is called the higher cut-off frequency. The gain of an amplifier without feedback can be represented in terms of this higher cut-off frequency as [Equ.(10-)],

$$A_H = \frac{A_m}{1 + j(f/f_H)} \quad (13-10)$$

Where,  $A_m$  is the mid-frequency voltage gain and  $f_H$  is the lower cut-off frequency.

With negative feedback, the voltage gain in terms of lower cut-off frequency can be represented as,

$$A'_H = \frac{A_H}{1 + \beta A_H} \quad (13-11)$$

Putting the value of  $A_H$  from Equ.(13-9) we get,

$$A'_L = \frac{A_m / [1 + j f / f_H]}{1 + \beta A_m / [1 + j f / f_H]}$$

or,

$$A'_L = \frac{A_m}{1 + j f / f_H + \beta A_m}$$

or,

$$A'_L = \frac{A_m}{1 + \beta A_m} \times \frac{1}{1 + j f / f_H (1 + \beta A_m)}$$

Using Equ.(13-9),

$$A'_L = \frac{A'_m}{1 + j f / f'_H}$$

where,

$$\boxed{f'_H = f_H (1 + \beta A_m)} \quad (13-12)$$

From Equ.(13-10), it is clear that the lower cut-off frequency has been increased, due to the negative feedback, by a factor  $(1 + \beta A_m)$ .

### 13.8 Effect of Negative Feedback on Bandwidth

The range of frequencies which is properly amplified by an amplifier is called its bandwidth. It is defined by the difference between the higher and lower cut-off frequencies. The bandwidth of an amplifier without feedback will be,

$$BW = f_H - f_L$$

where,  $f_H$ , and  $f_L$  are the higher and lower cut-off frequencies without feedback. When negative feedback is employed in an amplifier the lower cut-off frequency decrease and the higher cut-off frequency increases. Then the bandwidth with negative feedback will be,

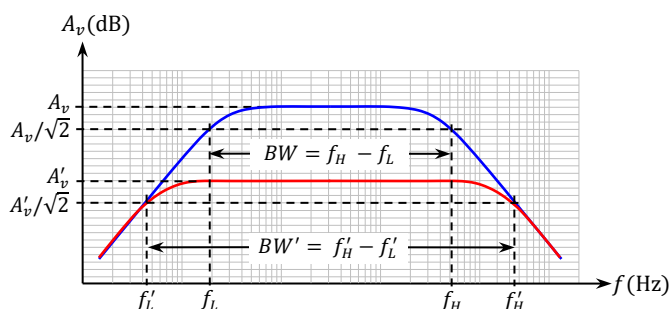


Fig.13-4: The bandwidth of an amplifier with and without feedback

$$BW' = f'_H - f'_L \quad (13-13)$$

where,  $f'_H$ , and  $f'_L$  are the higher and lower cut-off frequencies with feedback. From Equ.(13-8) and Equ.(13-11) we find that  $f'_H > f_H$ , and  $f'_L < f_L$ . Therefore, we can say that  $BW' > BW$ . That is the negative feedback has increased the bandwidth of the amplifier. The situation is illustrated in Fig.13-3. Negative feedback increased the bandwidth, but reduced the mid-band gain.

#### Example 13-3

An amplifier has an open-loop mid-band gain of 200. The lower and higher cut-off frequencies of the amplifier are 100 Hz and 200 kHz, respectively. A negative feedback with  $\beta = 0.1$  is applied to the amplifier. Calculate the bandwidth of the amplifier with and without feedback.

#### Solution:

Bandwidth of the amplifier without feedback will be,

$$BW = f_H - f_L = 200 \text{ kHz} - 0.2 \text{ kHz} = 199.8 \text{ kHz [Ans.]}$$

With negative feedback the lower cut-off frequency will be,

$$\text{Using Equ.(13-8)} \quad f'_L = \frac{f_L}{1 + \beta A_m} = \frac{100 \text{ Hz}}{1 + (0.1)200} = 4.8 \text{ Hz}$$

With negative feedback the higher cut-off frequency will be,

$$\text{Using Equ.(13-11)} \quad f'_H = f_H(1 + \beta A_m) = 200 \text{ kHz}(1 + 0.1 \times 200) = 4200 \text{ kHz}$$

Therefore, the bandwidth with feedback will be,

Using Equ.(13-13)  $BW' = 4200 \text{ kHz} - 4.8 \text{ Hz} \approx 4200 \text{ kHz}$  [Ans.]

**Comments:** Bandwidth has been increased by a factor  $(1 + A_m\beta) = 21$ .

### 13.9 Decreased Noise

When an amplifier amplifies a signal, it also amplifies noises from various sources. It may be thermal noise or shot noise (due to random movement of electrons and holes), miscellaneous noise, etc. If negative feedback is used in an amplifier the noise voltage (Let  $N$ ) will be reduced by a factor  $1/(1 + \beta A)$  in the similar manner to non-linear distortion. So the value of noise in an amplifier with feedback will be,

$$N' = \frac{N}{1 + \beta A}$$

However, if the gain is increased by adding additional stages of amplifiers, the noise will be increased again. If the increase in gain can be achieved merely by adjustment of circuit parameters, we can get definite reduction in noise with the help of negative feedback.

### 13.10 Effect of Negative Feedback on Input and Output Impedance

Negative feedback increases input impedance and decreases output impedance. However, this effect depends on the topologies of the feedback circuits. In the next section we will find that some topologies increases input impedance and decreases output impedance while some other topologies decrease input impedance and increases output impedance. But in both cases negative feedback improve the performance of the amplifier in the context of input and output impedance.

### 13.11 Feedback Amplifier Topologies

The feedback signal may be proportional to the output voltage or output current. Similarly the feedback signal itself may be either a voltage signal or a current signal. The combinations of these options results in four types of topologies of designing feedback amplifiers. These are:

1. Voltage Series Feedback Amplifier
2. Voltage Shunt Feedback Amplifier
3. Current Series Feedback Amplifier
4. Current Shunt Feedback Amplifier

In the names of topologies of feedback amplifiers, the first term (either voltage or current) indicates what output parameter (voltage/current) the feedback signal is proportional to. The second term of the names (series or shunt) means how the feedback signal is connected to the source signal. For example, **voltage series feedback** means the feedback signal is proportional to output voltage and the feedback signal is connected in series with source signal [Fig.13-3(a)]. Voltage series feedback is also called **voltage sampling voltage mixing feedback**. In the same way, **current shunt feedback** (also called **current sampling current mixing feedback**) means the feedback signal is proportional to output current and it is connected in parallel to the source signal [Fig.13-3(b)]. To generate a

feedback signal proportional to output voltage, the whole output voltage has to be applied to the input of feedback network, that is the two terminals of the feedback network have to be connected in parallel with load resistance,  $R_L$  [Fig.13-3(a, b)]. Similarly, To generate a feedback signal proportional to output current, the whole output current has to be applied to the feedback network, that is the input side of feedback network has to be connected in series with load resistance,  $R_L$  [Fig.13-3(c, d)]. Another point to note that if the feedback signal is connected in series with the source signal, the source signal and the feedback signal must be voltage signal. Similarly, if the feedback signal is connected in parallel to the source, the source signal and the feedback signal both must be current signal. For this reason, the amplifiers of Fig.13-3(a) and Fig.13-3(c) takes voltage as input and in case of Fig.13-3(b) and Fig.13-3(d) the amplifiers are taking current as input signal. In this context, the amplifiers in Fig.13-3(a) and Fig.13-3(c) are **voltage amplifiers** or **trans-conductance amplifiers**, and the amplifiers in Fig.13-3(b) and Fig.13-3(d) are **current amplifiers** or **trans-resistance amplifiers**.

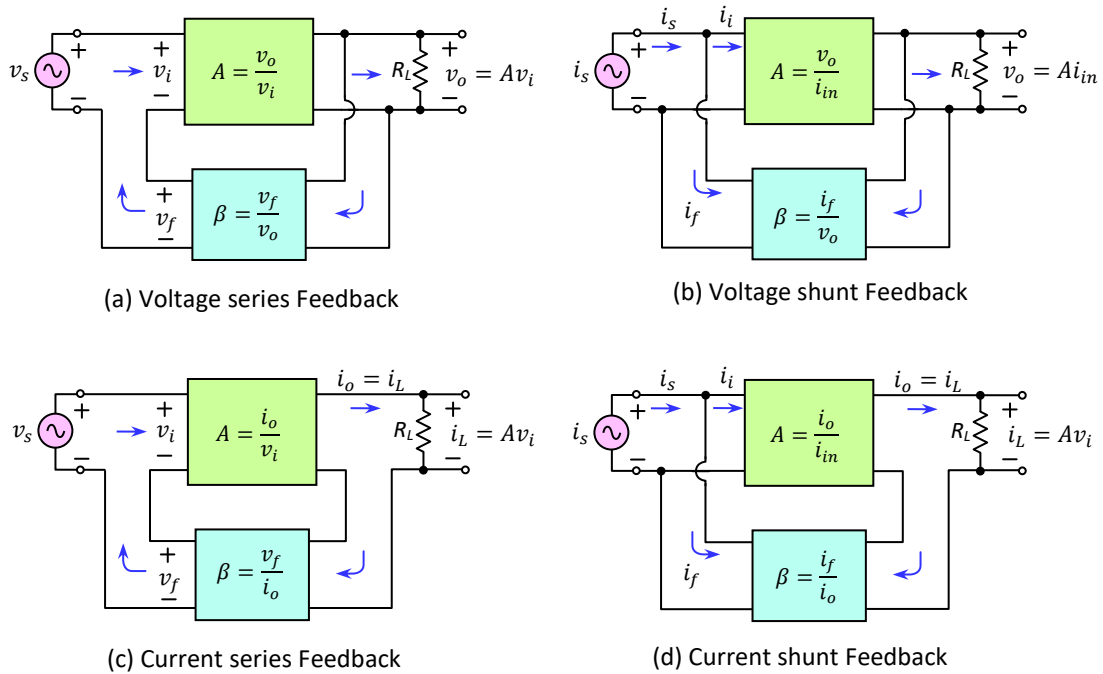


Fig.13-5: Four topologies of feedback amplifier circuits

As shown in Fig.13-3(a) and Fig.13-3(c), the polarity of  $v_f$  is such that the overall input to the amplifier is lower than  $v_s$ . By applying KVL, it can easily be found that  $v_i = v_s - v_f$ . Thus, the feedback used in the circuits of Fig.13-3(a) and Fig.13-3(c) are negative feedback. **We would have got positive feedback, if the polarity of feedback voltage  $v_f$  were just opposite.** Similarly, the direction of  $i_f$  in Fig.13-3(b) and Fig.13-3(d), is such that the overall input current to the amplifier is lower than  $i_s$ . By applying KCL, it can easily be found that  $i_i = i_s - i_f$ . Thus, the feedbacks used in these circuits are negative feedback. If the direction of  $i_f$  were opposite (outward from feedback network), the

feedback were positive feedback. Note that, in case of AC signal, the polarity/direction of input signal indicates the situation for positive half cycles.

### 13.12 Voltage Series Feedback

Fig.13-6 shows the voltage series feedback topology. Here, the feedback signal ( $v_f$ ) is proportional to output voltage, that's why full output voltage is applied to the feedback network. The feedback voltage ( $v_f$ ) is connected in series with the source voltage ( $v_s$ ). As  $v_f$  is subtracted from the source voltage ( $v_s$ ), the arrangement is a negative feedback. Here,  $A$  is the voltage gain of the amplifier without feedback and when no load is connected to the output terminals. The voltage gain will be reduced due to source internal resistance ( $r_s$ ) and load resistance ( $R_L$ ).

#### Voltage Gain

If there is no feedback ( $v_f = 0$ ), then  $v_i = v_s - v_f = v_s$ . If the output terminal is open or the load is not connected to output then  $v_o = Av_i$ . In that case, the voltage gain of the amplifier will be,

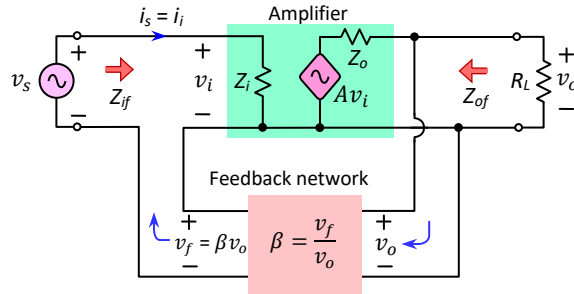


Fig.13-6: Voltage series feedback amplifier circuit

$$A = \frac{v_o}{v_s} = \frac{v_o}{v_i}$$

If we connect the load  $R_L$ , the output voltage will be,

$$v_o = \frac{Av_i \times R_L}{R_L + Z_o}$$

Then the voltage gain of the amplifier will be,

$$A' = \frac{v_o}{v_i} = \frac{A \times R_L}{R_L + Z_o}$$

If negative feedback is used in the amplifier, then  $v_i = v_s - v_f$ . So the output voltage will be,

$$v_o = A'v_i = A'(v_s - v_f)$$

We know that,  $\beta = v_f/v_o$ . Therefore,  $v_f = \beta v_o$ . Putting this value into the above equation, we get,

$$v_o = A'(v_s - \beta v_o) = A'v_s - A'\beta v_o$$

$$\text{or,} \quad v_o + A'\beta v_o = A'v_s$$

$$\text{or,} \quad v_o(1 + A'\beta) = A'v_s$$

$$A_f = \frac{v_o}{v_s} = \frac{A'}{1 + A'\beta} \quad (13-14)$$

Here,  $A_f$  is the gain of the amplifier with feedback. In case of positive feedback the gain will be,

$$A_f = \frac{A'}{1 - A'\beta} \quad (13-15)$$

Remember that here,  $A$  is the voltage gain without feedback and no load,  $A'$  is the voltage gain without feedback but with load and  $A_f$  is the voltage gain with feedback and load connected.

### Input Impedance

From Fig.13-6, without feedback we get  $v_i = v_s$ . Then the input impedance will be,

$$Z_i = \frac{v_s}{i_s} = \frac{v_i}{i_i}$$

When the feedback is used, the input impedance will be,

$$Z_{if} = \frac{v_s}{i_s} \quad (13-16)$$

From Fig.13-6, without feedback we get,

$$\begin{aligned} v_s &= i_s Z_i + \beta v_o \\ \therefore v_o &= v_i A' & v_s &= i_s Z_i + \beta(v_i A') \\ \therefore & & v_s &= i_s Z_i + \beta(i_s Z_i A') \end{aligned}$$

Putting this value into Equ.(13-15) we get,

$$Z_{if} = \frac{i_s Z_i + \beta(i_s Z_i A')}{i_s}$$

$$\text{or,} \quad Z_{if} = Z_i + \beta(Z_i A')$$

$$\therefore \quad \boxed{Z_{if} = Z_i(1 + \beta A')} \quad (13-17)$$

From this expression we find that due to the negative feedback, input impedance has been increased by a factor  $(1 + \beta A')$ .

### Output Impedance

To determine the output impedance, we have to short the input source, that is,  $v_s = 0$ . Then we have to apply a suitable voltage, say,  $v$  at the output and determine the output current ( $i$ ). In that case using KVL, we can write,

$$v = iZ_o + Av_i$$

As  $v_s = 0$ . Therefore,  $v_i = -v_f = -\beta v$ . Putting this value into the above equation,

$$v = iZ_o + A(-\beta v)$$

$$\text{or,} \quad v + A\beta v = iZ_o$$

$$\therefore v(1 + A\beta) = iZ_o$$

$$\therefore Z_{of} = \frac{v}{i} = \frac{Z_o}{1 + A\beta} \quad (13-18)$$

From this expression we can say that, due to the negative feedback, output impedance has been decreased by a factor  $(1 + A\beta)$ . Here,  $A$  is the open circuit gain of the amplifier.

### 13.13 Voltage Shunt Feedback

Fig.13-7 shows the voltage shunt feedback topology. Here, the feedback is a current signal ( $i_f$ ) which is proportional to output voltage. That is, the feedback network samples the output voltage. The feedback current ( $i_f$ ) is connected in parallel with the input source. As  $i_f$  is subtracted from the source current ( $i_s$ ), the arrangement is negative feedback. The amplifier is a trans-resistance amplifier that takes current as input and produces voltage at the output. Here,  $A$  is the gain of the amplifier without feedback and without load ( $R_L$ ), that is,  $A = v_o/i_i$ .

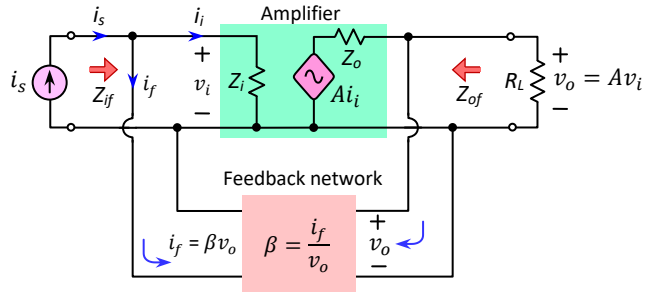


Fig.13-7: Voltage shunt feedback amplifier

#### Gain of the Amplifier

The gain of the amplifier without feedback and no load will be,

$$A = \frac{v_o}{i_i} = \frac{v_o}{i_s}$$

The gain of the amplifier without feedback but with load will be,

$$A' = \frac{AR_L}{Z_o + R_L} \quad (13-19)$$

When feedback is applied to this circuit, using KCL, we get  $i_i = i_s - i_f$ . Then the output voltage will be,

$$v_o = \frac{(A i_{in}) R_L}{Z_o + R_L} = (i_s - i_f) A'$$

$$\text{or,} \quad v_o = (i_s - \beta v_o) A'$$

$$\therefore v_o(1 + A'\beta) = i_s A'$$

So, the gain of the amplifier with feedback will be,

$$A_f = \frac{v_o}{i_s} = \frac{A'}{1 + A'\beta} \quad (13-20)$$

This equation shows that the gain of the amplifier with feedback has been decreased by a factor  $1/(1 + A'\beta)$ .

### Input Impedance

As shown in Fig.13-7, without feedback  $i_i = i_s$ . Then the input impedance will be,

$$Z_i = \frac{v_i}{i_i} \quad \text{or,} \quad v_i = Z_i i_i$$

When the feedback is used, the input impedance will be,

$$Z_{if} = \frac{v_i}{i_s} = \frac{i_i Z_i}{i_i + i_f}$$

But  $i_i = i_s - i_f$ . Therefore,

$$Z_{if} = \frac{i_{in} Z_i}{i_{in} + \beta v_o}$$

$$\text{or,} \quad Z_{if} = \frac{v_i}{i_{in} + \beta i_{in} A'}$$

$$\text{or,} \quad Z_{if} = \frac{v_i / i_{in}}{1 + \beta A'}$$

$$\therefore \quad \boxed{Z_{if} = \frac{Z_i}{1 + \beta A'}} \quad (13-21)$$

From this expression we find that due to the negative feedback, input impedance has been decreased by a factor  $1/(1 + \beta A')$ . **For trans-resistance amplifier the lower is the input impedance the better is the amplifier.**

### Output Impedance with Feedback

The output circuit of this amplifier is same as that of voltage series feedback. Therefore, the output impedance with feedback will be same. That is,

$$\boxed{Z_{of} = \frac{Z_o}{1 + \beta A}} \quad (13-22)$$

#### 13.14 Current Series Feedback

Fig.13-8 shows the current series feedback topology. Here, the feedback signal ( $v_f$ ) is proportional to output current, that's why full output current passes through the feedback network. The feedback voltage ( $v_f$ ) is connected in series with the source voltage ( $v_s$ ). As  $v_f$  is out of phase from the source voltage ( $v_s$ ), this arrangement is a negative feedback. Here, the amplifier is a

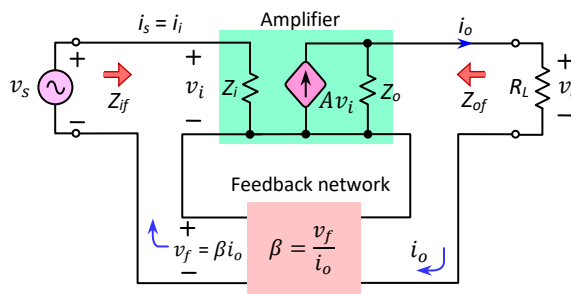


Fig.13-8: Current series feedback amplifier

transconductance amplifier and  $A$  is the gain of the amplifier without feedback and with  $R_L = 0$  (output terminals shorted).

### Voltage Gain

If there is no feedback,  $v_i = v_s$  and if the output terminals are short, then  $i_o = Av_i$ . In that case, the gain of the amplifier will be,

$$A = \frac{i_o}{v_s} = \frac{i_o}{v_i}$$

But if the load is connected at the output, then

$$i_o = \frac{Av_i Z_o}{Z_o + R_L} = v_i A' \quad (13-23)$$

Then the gain of the amplifier will be,

$$A' = \frac{i_o}{v_i} = \frac{1}{v_i} \times \frac{Av_i Z_o}{Z_o + R_L} = \frac{AZ_o}{Z_o + R_L}$$

If the negative feedback is employed in the circuit,

$$v_i = v_s - v_f = v_s - \beta i_o$$

Putting this value of  $v_i$  into Equ.(13-22)

$$i_o = (v_s - \beta i_o)A'$$

or,

$$i_o + \beta i_o A' = v_s A'$$

$\therefore$

$$i_o(1 + \beta A') = v_s A'$$

Therefore, gain of the amplifier with negative feedback (with load connected) will be,

$$A_f = \frac{i_o}{v_s} = \frac{A'}{1 + \beta A'} \quad (13-24)$$

Like the other topologies, current series negative feedback also reduces the gain by a factor  $1/(1 + \beta A')$ .

For positive feedback the gain of the amplifier will be,

$$A_f = \frac{i_o}{v_s} = \frac{A'}{1 - \beta A'} \quad (13-25)$$

### Input Impedance with Feedback

The input circuit of this amplifier is same as that of voltage series feedback. Therefore, the input impedance with feedback will be same. That is,

$$Z_{if} = Z_i(1 + \beta A) \quad (13-26)$$

### Output Impedance

To determine the output impedance, we have to short the input source, that is,  $v_s = 0$ . Then we have to replace  $R_L$  by a suitable current, say,  $i$ . Let due to this circuit the output voltage be  $v$ . Using KCL, we can write,

$$i = Av_i + \frac{v}{Z_o}$$

As  $v_s = 0$ . Therefore,  $v_i = -v_f = -\beta i$ . Putting this value into the above equation,

$$i = -A\beta i + \frac{v}{Z_o}$$

$$\text{or,} \quad i + A\beta i = \frac{v}{Z_o}$$

$$\text{or,} \quad i(1 + A\beta) = \frac{v}{Z_o}$$

$$\therefore \quad \frac{v}{i} = Z_o(1 + A\beta) = Z_{of}$$

Therefore, the output impedance with feedback will be,

$$\boxed{Z_{of} = Z_o(1 + A\beta)} \quad (13-27)$$

From this expression we can say that due to the negative feedback, output impedance has been increased by a factor  $(1 + A\beta)$ . Here,  $A$  is the open circuit gain of the amplifier. For a transconductance amplifier, the higher is the output impedance the better is the amplifier.

### 13.15 Current Shunt Feedback

Fig.13-9 shows the current shunt feedback topology. Here, the feedback signal ( $i_f$ ) is proportional to output current, that's why full output current passes through the feedback network. The feedback current ( $i_f$ ) is connected in parallel to the source ( $i_s$ ). The direction of  $i_f$  is such that it decreases the value of  $i_i$ . So, this arrangement is a negative feedback. Here, the amplifier is a current amplifier and  $A$  is the gain of the amplifier without feedback and with  $R_L = 0$  (output terminals shorted).

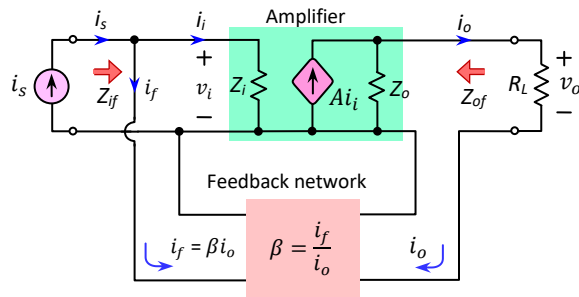


Fig.13-9: Current shunt feedback amplifier circuit

### Voltage Gain

If there is no feedback, then  $i_i = i_s$  and if the load ( $R_L$ ) is short,  $i_o = Ai_i$ . Then the gain of the amplifier will be,

$$A = \frac{i_o}{i_s} = \frac{i_o}{v_i}$$

But if the load is connected at the output, then

$$i_o = \frac{A i_i Z_o}{Z_o + R_L} = i_i A' \quad (13-28)$$

Then the gain of the amplifier will be,

$$A' = \frac{i_o}{i_s} = \frac{i_o}{i_i} = \frac{1}{i_i} \times \frac{A i_i Z_o}{Z_o + R_L} = \frac{A Z_o}{Z_o + R_L}$$

If the negative feedback is employed in the circuit,

$$i_i = i_s - i_f = i_s - \beta i_o$$

Putting this value of  $i_i$  into Equ.(13-27),

$$i_o = i_i A' = (i_s - \beta i_o) A'$$

or,

$$i_o + \beta i_o A' = i_s A'$$

$\therefore$

$$i_o (1 + \beta A') = i_s A'$$

Therefore, gain of the amplifier with negative feedback (with load connected) will be,

$$A_f = \frac{i_o}{i_s} = \frac{A'}{1 + \beta A'} \quad (13-29)$$

Like the other topologies, current series negative feedback also reduces the gain by a factor  $1/(1 + \beta A')$ .

For positive feedback the gain of the amplifier will be,

$$A_f = \frac{i_o}{i_s} = \frac{A'}{1 - \beta A'} \quad (13-30)$$

### Input Impedance with Feedback

The input circuit of this amplifier is same as that of voltage shunt feedback. Therefore, the input impedance with feedback will be same. That is,

$$Z_{if} = \frac{Z_i}{1 + A\beta} \quad (13-31)$$

### Output Impedance with Feedback

The output circuit of this amplifier is same as that of current series feedback. Therefore, the output impedance with feedback will increase. That is,

$$Z_{of} = Z_o (1 + A\beta) \quad (13-32)$$

## 13.16 Comparison of Feedback Topologies

All topologies with negative feedback decrease gain. Some topologies increase output impedance but decrease output impedance. On the contrary, there are some topologies that decrease input

impedance and increases output impedance. All the topologies are compared in Table-1 considering negative feedback.

Table 13-1: Comparison of four feedback topologies

| Characteristics  | Types of topologies |               |                |               |
|------------------|---------------------|---------------|----------------|---------------|
|                  | Voltage Series      | Voltage Shunt | Current Series | Current Shunt |
| Voltage gain     | Decrease            | Decrease      | Decrease       | Decrease      |
| Input impedance  | Increase            | Decrease      | Increase       | Decrease      |
| Output impedance | Decrease            | Decrease      | Increase       | Increase      |
| Bandwidth        | Increase            | Increase      | Increase       | Increase      |
| Distortion       | Decrease            | Decrease      | Decrease       | Decrease      |
| Noise            | Decrease            | Decrease      | Decrease       | Decrease      |

### 13.17 Practical Feedback Circuits

Negative feedback provides many advantages at the cost of reducing gain. So, the amplifiers having high gain employ negative feedback to improve its performance by sacrificing some gain. When amplifiers are used as multivibrators (discussed in Chapter) or in oscillators (discussed in Chapter), positive feedback is used. In this topic, we will discuss only negative feedback. In practical amplifiers any of the four topologies of feedback can be used. Sometimes, we cannot easily understand the topology of the feedback used in practical amplifiers. The feedback signal ( $v_f$  or  $i_f$ ) may be proportional to the output voltage or output current. If the feedback network (generally a single resistor or a voltage divider network) is connected to the same terminal from where the output voltage is taken, then it is voltage sampling feedback. To understand this, we can short the output load so that  $v_o = 0$ . If the feedback signal becomes zero, due to this short circuit, then it is voltage sampling feedback. Similarly, if the feedback network is connected to a different terminal, then it is current sampling feedback. To understand this, we can open the output load to make  $i_o = 0$ . In that case, the feedback signal will be zero. On the contrary, the voltage sensing feedback will increase if the output terminal is open, and the current sensing feedback sign will increase if the output terminals are shorted.

In case of voltage mixing, the feedback signal will be a voltage signal ( $v_f$ ) and in case of current mixing the feedback signal will be a current signal ( $i_f$ ). If the feedback signal ( $i_f$ ) and the source signal ( $v_f$  or  $i_f$ ) are connected to the same node, it is called current mixing feedback, and if the feedback signal ( $v_f$ ) and the source signal ( $v_f$  or  $i_f$ ) are connected to different nodes, it will be voltage mixing feedback. We have found before, voltage series feedback is also called voltage sampling and series mixing feedback or simply parallel-series feedback.

### 13.18 Practical Circuit of Voltage Series Feedback

There are many amplifier circuits that employ voltage series feedback. In the following topics some practical voltage series feedback amplifiers are discussed.

#### 13.18.1 Voltage Series Feedback in CS Amplifier

Fig.13-10 shows a common source FET amplifier where voltage series feedback has been used. The output voltage has been divided by a voltage divider network consisting of  $R_1$  and  $R_2$ . The feedback voltage will be,

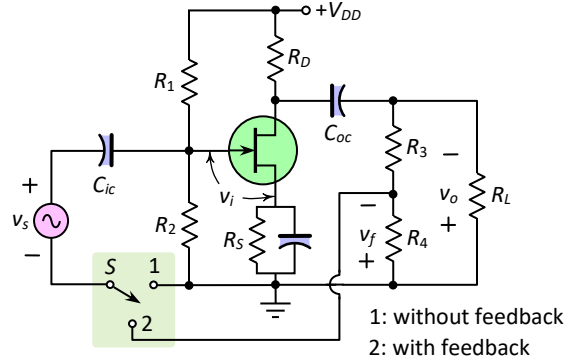


Fig.13-10: CS amplifier with voltage series feedback

$$v_f = -v_o \frac{R_4}{R_3 + R_4}$$

or,

$$\beta = \frac{v_f}{v_o} = -\frac{R_4}{R_3 + R_4} \quad (13-33)$$

When the switch is connected to position '1' no feedback will be effective. We know that in the mid-band frequencies, all the external capacitors will work as short circuits. Therefore, we can write  $v_i = v_s$ , that is, there is no feedback. The voltage gain of this amplifier without feedback will be,

$$A_v = -g_m R_L \quad (13-34)$$

where,

$$R_L = R_D \parallel (R_3 + R_4) \parallel R_L \quad (13-35)$$

The input impedance of the amplifier without feedback is,

$$Z_i \approx R_1 \parallel R_2 \quad (13-36)$$

And output impedance without feedback is,

$$Z_o \approx R_D \quad (13-37)$$

When the switch is connected to position '2', a feedback will be effective in the circuit. From the circuit we can write,  $v_i = v_s - v_f$  (consider all external capacitors short). As  $v_i$  is decreased due to feedback, it is negative feedback.

Voltage gain of the amplifier with feedback will be,

$$A_{vf} = \frac{A_v}{1 + A_v \beta} = \frac{-g_m R_L}{1 + g_m R_L [R_2 / (R_1 + R_2)]} \quad (13-38)$$

If  $g_m R_L [R_2 / (R_1 + R_2)] \gg 1$ , then

$$A_{vf} = \frac{-g_m R_L}{g_m R_L [R_2 / (R_1 + R_2)]} = -\frac{R_1 + R_2}{R_2}$$

$$\therefore \boxed{A_{vf} = -\frac{1}{\beta}} \quad (13-39)$$

The input impedance of the amplifier with feedback is,

$$\boxed{Z_{if} \approx R_1 \parallel R_2 (1 + A_v \beta)} \quad (13-40)$$

And output impedance with feedback is,

$$\boxed{Z_o \approx \frac{R_D}{1 + A_v \beta}} \quad (13-41)$$

### Example 13-4

For the voltage series feedback of Fig.13-11, calculate voltage gain with and without feedback. Assume  $V_P = -3$  V, and  $I_{DSS} = 12$  mA for the FET.

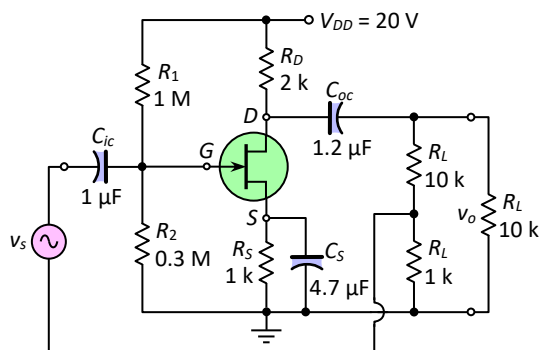


Fig.13-11: Voltage series feedback amplifier for Example 12-4

### Solution:

Using Equ.(7-24), 
$$V_G = V_{R2} = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{20 \text{ V} \times 300 \text{ k}}{1 \text{ M} + 300 \text{ k}} = 4.62 \text{ V}$$

Using Equ.(7-25), 
$$V_{GS} = V_G - I_D R_S = 4.62 \text{ V} - 1 \text{ k} \times I_D \text{ mA}$$

Using Equ.(7-76), 
$$I_D = I_{DSS} \left( 1 - \frac{|I_D R_S - V_G|}{|V_P|} \right)^2 = 12 \text{ mA} \left( 1 - \frac{I_D - 4.62 \text{ V}}{3 \text{ V}} \right)^2$$

Solving this equation for  $I_D$  we will get,  $I_D \approx 5.58$  mA.

Therefore, the gate-source voltage will be  $V_{GS} = V_G - I_D R_S = 4.62 \text{ V} - 5.58 \text{ mA} \times 1 \text{ k} = -0.96 \text{ V}$ .

The value of  $g_m$  at this operating point will be,

Using Equ.(6-6), 
$$g_m = \frac{2I_{DSS}}{|V_P|} \left( 1 - \left| \frac{V_{GS}}{V_P} \right| \right) = \frac{2 \times 12 \text{ mA}}{3 \text{ V}} \left( 1 - \frac{0.96 \text{ V}}{3 \text{ V}} \right) = 5.44 \text{ mS}$$

The gain of the amplifier without feedback will be,

Using Equ.(13-33), & Equ.(13-34)  $A_v = -g_m[R_D || R_L || (R_6 + R_7)]$

or,  $A_v = 5.44 \text{ mS} [2 \text{ k} || 10 \text{ k} || (10 \text{ k} + 1 \text{ k})] = 7.87 \text{ [Ans.]}$

From the circuit we find that the value of  $\beta$  will be,

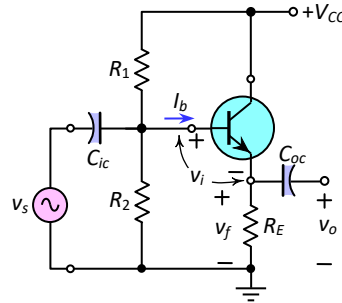
Using Equ.(13-32),  $\beta = \frac{R_7}{R_6 + R_7} = \frac{1 \text{ k}}{1 \text{ k} + 10 \text{ k}} = 0.091$

The gain of the amplifier with feedback will be,

Using Equ.(13-37),  $A_{vf} = \frac{A_v}{1 + A_v\beta} = \frac{7.87}{1 + 7.87 \times 0.091} \approx 4.59 \text{ [Ans.]}$

### 13.18.2 Voltage Series Feedback in CC Amplifier

Common collector amplifier (CC) or emitter follower is another example of voltage series feedback. Fig.13-12 shows a CC amplifier, where the output is taken from the emitter terminal and the input is applied to the base. The output voltage ( $v_o$ ) also works as the feedback voltage ( $v_f$ ). The feedback voltage is equal to output voltage and connected in series with  $v_s$ . Fig.13-12: Voltage series feedback in CC amplifier. Therefore, it is a voltage series feedback.



From the circuit, input voltage  $v_i$ , can be calculated as,

$$v_i = v_s - v_f$$

If we assume that there is no feedback, that is,  $v_f = 0$ , then

$$v_i = v_s - 0 = v_s$$

Then, the voltage gain of the amplifier without feedback can be calculated as,

$$A_v = \frac{v_o}{v_i} = \frac{v_o}{v_s} = \frac{i_e R_E}{v_s} \approx \frac{h_{fe} i_b R_E}{v_s} = \frac{h_{fe} (v_s / h_{ie}) R_E}{v_s}$$

$$A_v = \frac{h_{fe} R_E}{h_{ie}} \quad (13-42)$$

And the input impedance without feedback (imagine that the emitter is directly connected to ground) will be,

$$Z_i = R_B || h_{ie} \approx h_{ie} [\because R_B \gg h_{ie}] \quad (13-43)$$

Although it is a CC amplifier, we have used  $h_{ie}$  and  $h_{fe}$  instead of  $h_{ic}$  and  $h_{fc}$ . In Chapter 5, it is shown that  $h_{ic} \approx h_{ie}$  and  $h_{fc} \approx h_{fe}$ . Hybrid parameters for CE configuration are commonly available.

Now, let us consider the effect of feedback. The feedback factor can be determined as,

$$\beta = \frac{v_f}{v_o} = 1 [\because v_f = v_o] \quad (13-44)$$

Therefore, the gain of the amplifier with feedback will be,

$$A_{vf} = \frac{A_v}{1 + A_v\beta} = \frac{h_{fe}R_E/h_{ie}}{1 + h_{fe}R_E/h_{ie}}$$

$$\boxed{A_{vf} = \frac{h_{fe}R_E}{h_{ie} + h_{fe}R_E}} \quad (13-45)$$

The input impedance of the amplifier with feedback will be,

$$Z_{if} = Z_i(1 + A_v\beta) = h_{ie}(1 + h_{fe}R_E/h_{ie})$$

$$\boxed{Z_{if} = h_{ie} + h_{fe}R_E} \quad (13-46)$$

Output impedance with feedback ( $Z_{of}$ ) can be determined using Equ.(5-123) as,

$$Z_{of} = \frac{h_{ie}}{h_{fe}} \quad (13-47)$$

If the output impedance is calculated including  $R_E$ , then

$$Z_{of} = \frac{h_{ie}}{h_{fe}} \parallel R_E \quad (13-48)$$

As we know, the voltage gain of a CC amplifier is very low. This low gain is due to the high degree of feedback that ensures an excellent bandwidth and other advantages of CC amplifier. Its high input impedance and low output impedance makes it ideal as the last stage in a cascaded amplifier system.

### 13.18.3 Voltage Series Feedback in Two Stage CE Amplifier

A two stage CE amplifier with voltage series feedback is shown in Fig.13-3. The feedback network consists of  $R_5$  and  $R_6$ . The value of feedback factor is,

$$\beta = \frac{R_6}{R_5 + R_6} \quad (13-49)$$

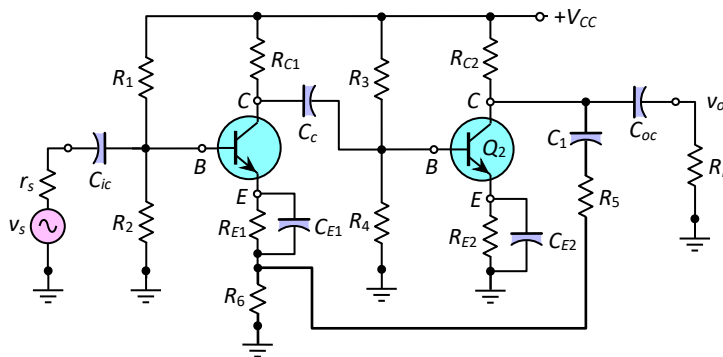


Fig.13-13: Voltage series feedback in two-stage CE amplifier

The overall voltage gain of the two stage amplifier is can be calculated using Equ.(11-19), as

$$A_v = \frac{h_{fe1}(R_{C1} \parallel R_{B2} \parallel h_{ie2})}{h_{ie1} + h_{fe1}R_6} \times \frac{h_{fe2}(R_{C2} \parallel R_L)}{h_{ie2}} \quad (13-50)$$

The voltage gain of this amplifier with feedback will be,

$$A_{vf} = \frac{A_v}{1 + \beta A_v} \quad (13-51)$$

The values of  $A_v$ , and  $\beta$  can be calculated using Equ.(13-46) and Equ.(13-47).

### Example 13-5

The direct coupled two-stage CE amplifier of Fig.13-14 utilizes voltage series feedback. Calculate the voltage gain with and without feedback for mid-frequency range. Assume for both transistors  $h_{fe} = 100$ .

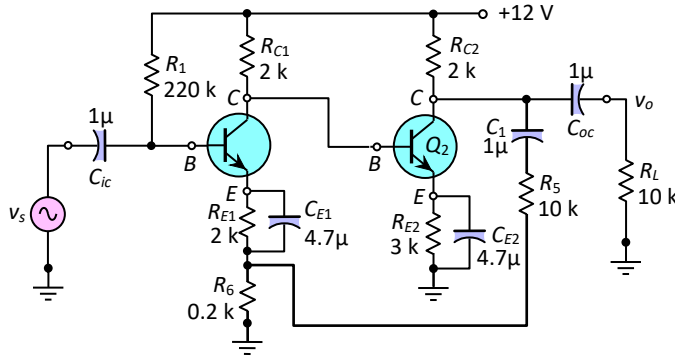


Fig.13-14: Voltage series feedback amplifier for Example 12-5

### Solution:

Using Equ.(4.21) 
$$I_{BQ1} = \frac{V_{CC} - V_{BE1}}{R_B + (\beta + 1)R_{E1}} = \frac{12 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega + (100 + 1) 2.2 \text{ k}\Omega} \approx 25.68 \text{ }\mu\text{A}$$

$$\therefore I_{CQ1} = I_{BQ1} \times h_{fe} = 25.68 \text{ }\mu\text{A} \times 100 = 2.568 \text{ mA}$$

$$\therefore V_{C1} = V_{CC} - I_{CQ1} \times R_{C1} = 12 \text{ V} - 2.568 \text{ mA} \times 2 \text{ k} \approx 6.86 \text{ V}$$

$$I_{EQ2} = \frac{V_{C1} - V_{BE2}}{R_{E2}} = \frac{6.86 \text{ V} - 0.7 \text{ V}}{3 \text{ k}\Omega} \approx 2.05 \text{ mA}$$

Therefore, 
$$h_{ie1} \approx h_{fe1} \frac{26 \text{ mV}}{I_{CQ1}} = 100 \times \frac{26 \text{ mV}}{2.568 \text{ mA}} \approx 1 \text{ k}$$

and, 
$$h_{ie2} \approx h_{fe2} \frac{26 \text{ mV}}{I_{EQ2}} = 100 \times \frac{26 \text{ mV}}{2.05 \text{ mA}} \approx 1.27 \text{ k}$$

Voltage gain without feedback can be calculated using Equ.(13-47),

$$A_v = \frac{100(2\text{ k} \parallel 1.27\text{ k})}{1\text{ k} + 100 \times 0.2\text{ k}} \times \frac{100(2\text{ k} \parallel 10\text{ k})}{1.27\text{ k}} \approx 485 \text{ [Ans.]}$$

Using Equ.(13.48),

$$\beta = \frac{R_6}{R_5 + R_6} = \frac{0.2\text{ k}}{10\text{ k} + 0.2\text{ k}} = \frac{1}{51}$$

Voltage gain without feedback can be calculated using Equ.(13-50),

$$A_{vf} = \frac{A_v}{1 + A_v\beta} = \frac{485}{1 + 485(1/51)} \approx 46 \text{ [Ans.]}$$

**Comments:** Due to negative feedback gain has decreased from 485 to 46.

### 13.18.4 Voltage Series Feedback in Op-Amp Non-inverting Amplifier

Another example of voltage series feedback is the op-amp non-inverting amplifier shown in Fig.13.15. The detail analysis of this circuit is given in **Chapter** . The feedback factor of this amplifier will be,

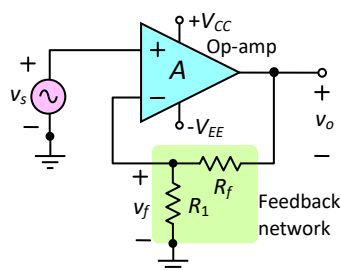


Fig.13-15: Voltage series feedback in non-inverting amplifier

$$\beta = \frac{v_f}{v_o} = \frac{R_1}{R_1 + R_f} \quad (13-52)$$

If the voltage gain of op-amp is  $A$ , the overall gain of the amplifier with feedback will be,

$$A_{vf} = \frac{A}{1 + A\beta} \approx \frac{A}{A\beta}$$

$$\therefore A_{vf} = \frac{1}{\beta} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1} \quad (13-53)$$

### 13.19 Voltage Shunt Feedback Amplifier

There are many amplifier circuits that employ voltage shunt feedback. In the following topics some of them are discussed.

### 13.19.1 Voltage Shunt Feedback in CE Amplifier

Fig.13-16 shows a CE amplifier with collector-to-base bias. This type of biasing technique is described in Section 4.14. This amplifier employs voltage shunt feedback through the resistor  $R_f$ . As discussed in Chapter 4.14, this negative feedback provides stability of the DC operating point. In the circuit of Fig.13-16,  $R_f$  provides negative feedback for AC signal, as the AC signal is not bypassed.

First of all we will try to understand the type of feedback that works in this circuit. We know that  $v_o$  is much larger than the input voltage  $v_i$ . So the direction of feedback current should be towards the base [opposite to the direction shown in Fig.13-17(a)]. But, as  $v_o$  is  $180^\circ$  out of phase from  $v_i$  this feedback current will be subtracted from  $i_s$  that ensures negative feedback.

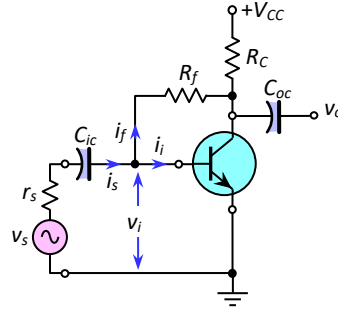


Fig.13-16: Voltage shunt feedback amplifier

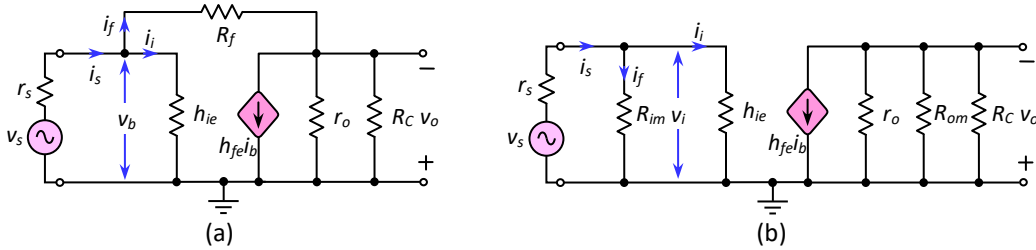


Fig.13-17: (a) Equivalent circuit of Fig.13-16, and (b) Same circuit using Miller's theorem

Thus, we can assume the direction of  $i_f$  from base to collector. The feedback signal is  $i_f$  is proportional to output voltage and mixes with  $i_s$ . Therefore, it is called *voltage sampling current mixing* or simply *voltage shunt feedback*. The value of feedback current will be,

$$i_f = \frac{v_i - v_o}{R_f} \approx -\frac{v_o}{R_f} [\because v_o \gg v_i]$$

$$\beta = \frac{i_f}{v_o} \approx -\frac{1}{R_f}$$

Fig.13-17(b) shows the equivalent circuit, where  $R_f$  has been split into an input resistance  $R_{im}$  and an output resistance  $R_{om}$  using Miller's theorem. If the gain of the amplifier is  $-A_{vf}$ , then,

$$R_{im} = \frac{R_f}{1 + A_{vf}} \quad (13-54)$$

And,

$$R_{om} = \frac{(-A_{vf})R_f}{-A_{vf} - 1} = \frac{A_{vf}R_f}{A_{vf} + 1} \approx R_f \quad (13-55)$$

### Input Impedance with Feedback

From Fig.13-17(b), we find that,

$$R_{if} = R_{im} \parallel h_{ie} \quad (13-56)$$

### Voltage Gain

The voltage gain with feedback is given by,

$$A_{vf} = \frac{v_o}{v_i} = -\frac{h_{fe}R'_L}{h_{ie}} \quad (13-57)$$

Considering the effect of source resistor, the voltage gain will be,

$$A_{vsf} = \frac{R_{if}}{r_s + R_{if}} A_{vf} \quad (13-58)$$

### Input Impedance

From Fig.13-17(b), it is clear that the input impedance with feedback will be,

$$Z_{if} = R_{im} \parallel h_{ie} \quad (13-59)$$

### Output Impedance

To calculate output impedance, let us replace the input source by its internal resistance  $r_s$  and apply a suitable voltage, say  $v_x$ , at the output terminal of Fig.13-17(a). If  $i_x$  is the output current for this voltage, then we can write

$$i_x = v_x \left[ \frac{1}{r_o} + \frac{1}{R_f + r_s \parallel h_{ie}} \right] + h_{fe} i_b \quad (13-60)$$

Using voltage divider rule, we can determine the base voltage as,

$$v_b = v_x \frac{r_s \parallel h_{ie}}{R_f + r_s \parallel h_{ie}} \quad (13-61)$$

Therefore, base current will be,

$$i_b = \frac{v_b}{h_{ie}} = \frac{v_x (r_s \parallel h_{ie})}{h_{ie} (R_f + r_s \parallel h_{ie})} \quad [\text{Using Equ.(13-61)}]$$

Putting this value of  $i_b$  into Equ.(13-60),

$$i_x = v_x \left[ \frac{1}{r_o} + \frac{1}{R_f + r_s \parallel h_{ie}} \right] + h_{fe} \frac{v_x (r_s \parallel h_{ie})}{h_{ie} (R_f + r_s \parallel h_{ie})}$$

From the above equation the output admittance with feedback will be,

$$y_{of} = \frac{i_x}{v_x} = \frac{1}{r_o} + \frac{1}{R_f + r_s \parallel h_{ie}} + \frac{h_{fe} (r_s \parallel h_{ie})}{h_{ie} (R_f + r_s \parallel h_{ie})} \quad (13-62)$$

If there were no feedback, the output admittance would have been

$$y_o = \frac{i_x}{v_x} = \frac{1}{r_o} \quad (13-63)$$

Comparing Equ.(13-62) and Equ.(13-63), we find that the output admittance with feedback has been increased. So the output impedance with feedback will decrease.

### Example 13-6

For the voltage shunt feedback amplifier of Fig.13-18, calculate overall voltage gain and input impedance. Assume all external capacitances are short circuit.

#### Solution:

Using Equ.(4.37),

$$I_{CQ} = \frac{V_{CC} - V_{BE}}{R_f/\beta + R_C} = \frac{6\text{ V} - 0.7\text{ V}}{82\text{ k}/50 + 2\text{ k}} \approx 1.46\text{ mA}$$

We know,  $r_e \approx \frac{26\text{ mV}}{I_{CQ}} = \frac{26\text{ mV}}{1.46\text{ mA}} = 17.81\ \Omega$

and,  $h_{ie} = r_e h_{fe} = 17.81\ \Omega \times 50 = 890\ \Omega$

The load resistance will be,

$$R'_L = R_C \parallel R_{om} = R_C \parallel R_f = 2\text{ k} \parallel 82\text{ k} = 1.95\text{ k}$$

Using Equ.(13-57),  $A_{vf} = -\frac{h_{fe} R'_L}{h_{ie}} = -\frac{50 \times 1.95\text{ k}}{890\ \Omega} = 109.56$

$$R_{im} = \frac{R_f}{1 + A_{vf}} = \frac{82\text{ k}}{1 + 109.56} \approx 0.74\text{ k}$$

$$R_{if} = R_{im} \parallel h_{ie} = 0.74\text{ k} \parallel 89\text{ k} = 0.41\text{ k} \text{ [Ans.]}$$

Overall gain will be

Using Equ.(13-58),  $A_{vsf} = \frac{R_{if} A_{vf}}{r_s + R_{if}} = \frac{0.41\text{ k} \times 109.56}{0.6\text{ k} + 0.41\text{ k}} \approx 44.48 \text{ [Ans.]}$

**Comments:** Due to the effect of  $r_s$  voltage gain has reduced from 109.56 to 44.48.

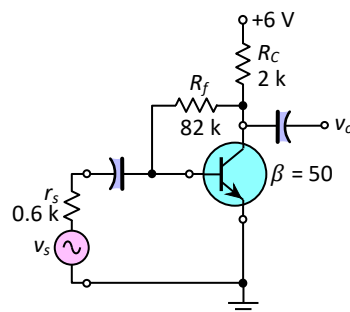


Fig.13-18: Voltage shunt feedback amplifier for Example 12-6

### 13.19.2 Voltage Shunt Feedback in Op-amp Inverting Amplifier

Another example of voltage shunt feedback is the op-amp inverting amplifier shown in Fig.13-19. The detail analysis of this circuit is covered in [Chapter](#) . The feedback factor of this amplifier is,

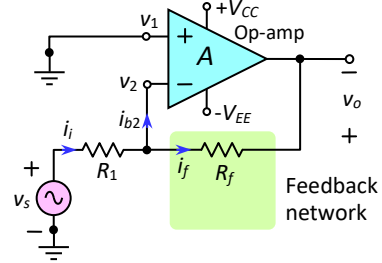


Fig.13-19: Voltage shunt feedback in inverting amplifier

$$i_f = \frac{v_o}{R_f}$$

$$\beta = \frac{i_f}{v_o} = \frac{1}{R_f}$$

(13- 64)

From the Fig.13-19, we can write,

$$i_i = i_f + i_{b2} \approx i_f$$

or,

$$\frac{v_s - v_2}{R_1} = \frac{v_2 - v_o}{R_f}$$

Due to virtual ground ([will be discussed in Chapter](#) ),  $v_2 = 0$ . Therefore,

$$\frac{v_s}{R_1} = \frac{-v_o}{R_f}$$

$$A_{vf} = \frac{v_{os}}{v_1} = -\frac{R_f}{R_1}$$

(13- 65)

### 13.20 Current Series Feedback

An example of current series feedback is a CE amplifier with an unbypassed emitter resistor. We can keep the whole  $R_E$  unbypassed or a portion of  $R_E$  unbypassed. Fig.13-20 shows a CE amplifier with an emitter  $R_f$ . A bypass capacitor is connected with a switch (s). If the switch is ON,  $R_f$  will be bypassed and there will have no feedback. If the switch is OFF,  $R_f$  will be unbypassed and negative feedback will be effective in the circuit.

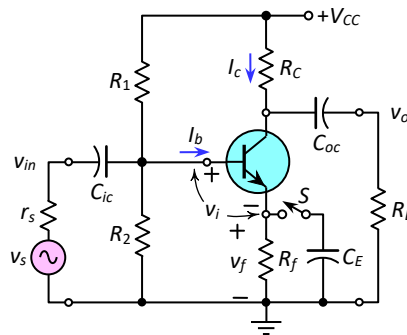


Fig.13-20: Current series feedback in CE amplifier

In Chapter 5, we have analyzed this circuit and found that, the voltage gain without feedback is,

$$A_v \approx -\frac{(R_C \parallel R_L)}{r_e} \quad (13-66)$$

If  $R_L$  is not connected, i.e., if  $R_L = \infty$   $A_v \approx -\frac{R_C}{r_e}$  (13-67)

In the same chapter, the voltage gain with unbypassed emitter resistor is calculated which is,

$$\therefore A_v \approx -\frac{(R_C \parallel R_L)}{r_e + R_f} \quad (13-68)$$

If  $R_L$  is not connected, i.e., if  $R_L = \infty$   $A_v \approx -\frac{R_C}{r_e + R_f}$  (13-69)

### Feedback Factor

Although it is current series feedback, but here, feedback factor will be calculated as  $v_f/v_o$ . Truly speaking,  $v_f = v_f/i_o$  does not produce correct result.

From the mid-frequency equivalent of Fig.13-20 we can find that (without considering  $R_L$ ),

$$v_o \approx -i_c R_C$$

And the value of feedback voltage is,

$$v_f = i_e R_f$$

Therefore, the feedback factor will be,

$$\beta = \frac{v_f}{v_o} = \frac{i_e R_f}{-i_c R_C} \approx -\frac{R_f}{R_C} \quad (13-70)$$

Using the theory of negative feedback, we can write,

$$A_{vf} = \frac{A_v}{1 + A_v \beta}$$

Using Equ.(13-67) and Equ.(13-70) into the above equation,

$$A_{vf} = \frac{-R_C/r_e}{1 + (-R_C/r_e)(-R_f/R_C)}$$

or,  $A_{vf} = \frac{-R_C/r_e}{1 + R_f/r_e}$

$$\therefore \boxed{A_{vf} = -\frac{R_C}{r_e + R_f}} \quad (13-71)$$

The voltage gain with feedback and load resistance will be,

$$\boxed{A_{vf} = -\frac{R_C \parallel R_L}{r_e + R_f}} \quad (13-72)$$

Using the feedback theory, we get the same voltage gain as found in Chapter 5 [Equ.(5-18)].

### Input Impedance

As shown in Chapter 5 [5-60], the input impedance of a CE amplifier with emitter bypass capacitor is give by,

$$Z_i = R_B || h_{ie} \approx h_{ie} = \beta r_e \quad (13-73)$$

The input impedance with feedback (with unbypassed emitter capacitor) can be calculated using Equ.(5-75),

$$Z_{if} = h_{ie} + (1 + h_{fe})R_f \approx h_{ie} + h_{fe}R_f \quad (5-75)$$

Let us calculate  $Z_{if}$  using the feedback theory. We know,

$$Z_{if} = Z_i(1 + A_v\beta)$$

Using Equ.(13-67), Equ.(13-70), and Equ.(13-73)

$$Z_{if} = h_{ie}[1 + (-R_C/r_e)(-R_f/R_C)]$$

or,

$$Z_{if} = h_{ie}[1 + R_f/r_e] = h_{ie} + h_{ie}R_f/r_e$$

$$\therefore h_{ie} \approx h_{fe}r_e$$

$$Z_{if} \approx h_{ie} + h_{fe}r_e R_f/r_e$$

$\therefore$

$$\boxed{Z_{if} \approx h_{ie} + h_{fe}R_f} \quad (13-74)$$

Equ.(5-75), and Equ.(13-75) Therefore, the input impedance calculated using also agrees with feedback theory.

### Example 13-7

For the current series feedback amplifier of Fig.13-21, determine the voltage gain and input impedance without and with feedback in the mid-frequency range.

#### Solution:

In determining the DC operating point we have to consider  $R_E = R_3 + R_f$ . As only  $R_f$  is kept unbypassed, this resistor will provide negative feedback.

To calculate  $r_e$  we need the DC operating point. For DC condition  $R_E = R_3 + R_f = 4.7 \text{ k}$ .

$$\text{Using Equ.(4-45),} \quad E_{Th} = \frac{V_{CC}R_2}{R_1 + R_2} = \frac{24 \text{ V} \times 24 \text{ k}}{33 \text{ k} + 24 \text{ k}} = 10.11 \text{ V}$$

$$\text{Using Equ.(4-46),} \quad E_{Th} = \frac{R_1R_2}{R_1 + R_2} = \frac{33 \text{ k} \times 24 \text{ k}}{33 \text{ k} + 24 \text{ k}} = 13.9 \text{ k}$$

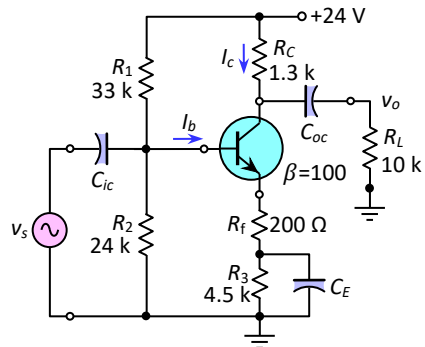


Fig.13-21: Current series feedback amplifier for Example 12-7

Using Equ.(4-49), 
$$I_C = \frac{\beta(E_{Th} - V_{BE})}{R_{Th} + (\beta + 1)R_E} = \frac{100(10.11 \text{ V} - 0.7 \text{ V})}{13.9 \text{ k} + (100 + 1)4.7 \text{ k}} \approx 1.93 \text{ mA}$$

Using Equ.(5-1), 
$$r_e \approx \frac{26 \text{ mV}}{I_E} \approx \frac{26 \text{ mV}}{I_C} = \frac{26 \text{ mV}}{1.93 \text{ mA}} \approx 13.5 \Omega$$

**Without feedback:**

Using Equ.(13-66), 
$$A_v = -\frac{R_C \parallel R_L}{r_e} = \frac{1.3 \text{ k} \parallel 10 \text{ k}}{13.5 \Omega} \approx 85$$

Using Equ.(13-73), 
$$Z_i \approx h_{ie} = \beta \times r_e = 100 \times 13.5 \Omega = 1.35 \text{ k}$$

**With feedback:**

Using Equ.(13-72), 
$$A_{vf} = -\frac{R_C \parallel R_L}{r_e + R_3} = \frac{1.3 \text{ k} \parallel 10 \text{ k}}{13.5 \Omega + 200 \Omega} \approx 5.4$$

Using Equ.(13-75), 
$$Z_{if} \approx h_{ie} + h_{fe}R_f = 13.5 \Omega + 100 \times 200 \Omega \approx 20 \text{ k}$$

**Comments:** A small portion of  $R_E$  is kept unbypassed. **If whole  $R_E$  was kept unbypassed, the gain would have drastically fallen.**

### 13.21 Current Shunt Feedback

Current shunt feedback is also called **current sampling and current mixing feedback** since a signal proportional to output current is mixed with input signal. The feedback signal is connected in parallel with the input source. Fig13-22 shows a two stage CE amplifier with current shunt feedback. The first stage is a CE amplifier with voltage divider bias and the second stage is a direct coupled CE amplifier that is biased by the collector voltage of first stage. Feedback is applied from the emitter of the second stage to the base of first stage through resistor  $R_f$ . Here, the feedback current  $i_f$  will be proportional to  $i_{e2}$  or the output current ( $\because i_{e2} \approx i_{c2}$ ). For the positive half-cycle of input signal,  $v_{c1}$  decreases and hence  $i_{e2}$  decreases. Therefore,  $i_f$  decreases. For the negative half-cycle of input,  $v_{c1}$  increases and  $i_{e2}$  increases also. Therefore,  $i_f$  increases. Thus,  $i_f$  is  $180^\circ$  out of phase with input current  $i_i$  which ensures negative feedback.

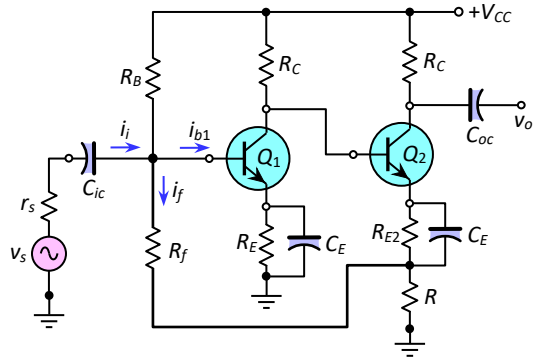


Fig.13-22: Current shunt feedback amplifier

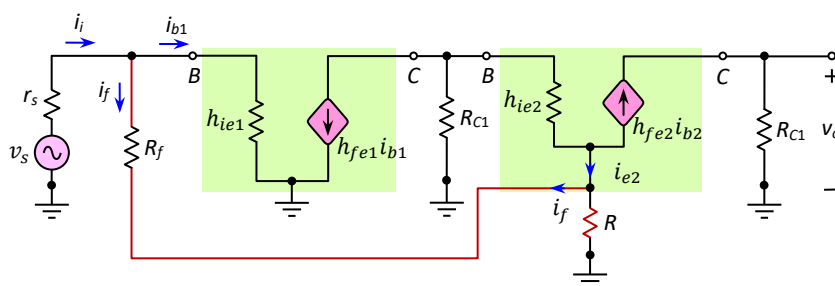


Fig.13-23: Equivalent circuit of Fig.12-22

From Fig.13-, it is observed that  $i_{e2}$  ( $= i_o$ ) divides between  $R_f$  and  $R$ . Using current divider rule we can write,

$$i_f = \frac{i_{e2}R}{R + (R_f + h_{ie1} \parallel r_s)} \approx \frac{i_o R}{R + (R_f + h_{ie1} \parallel r_s)}$$

$$\text{If } R_f \gg h_{ie1} \parallel r_s \quad i_f = \frac{i_o R}{R + R_f}$$

$$\therefore \quad i_f = \frac{i_o R}{R + R_f}$$

Therefore, the feedback factor will be,

$$\beta = \frac{i_f}{i_o} = \frac{R}{R + R_f} \quad (13-75)$$

### Current Gain without Feedback

From Fig.13-23, we find that collector current  $h_{fe1}i_{b1}$  will divide between two paths –  $R_{C1}$  and  $h_{ie2}$ . If  $R$  is very small, using current divider rule we can write,

$$i_{b2} = \frac{h_{fe1}i_{b1}R_{C1}}{R_{C1} + h_{ie2}} \quad (13-76)$$

The output current will be,

$$i_o \approx i_{c2} = h_{fe2}i_{b2}$$

Putting the value of  $i_{b2}$  from Equ.(13-76) into the above equation,

$$i_o = h_{fe2} \frac{h_{fe1}i_{b1}R_{C1}}{R_{C1} + h_{ie2}}$$

Therefore, the current gain will be,

$$A_i = \frac{i_o}{i_i} = \frac{i_o}{i_{b1}} = \frac{h_{fe1}h_{fe2}R_{C1}}{R_{C1} + h_{ie2}} \quad (13-77)$$

### Current Gain with Feedback

Using the theory of negative feedback, current gain with feedback can be calculated as,

$$A_{if} = \frac{A_i}{1 + A_i\beta} \quad (13-78)$$

where,  $A_i$  is given by Equ.(13-77) and  $\beta$  is given by Equ.(13-75).

### Input Impedance

If  $R_f$  is not connected, then the input impedance without feedback will be,

$$Z_i = h_{ie1}$$

As for shunt feedback input impedance decreases, the input impedance with feedback will be,

$$Z_{if} = \frac{h_{ie1}}{1 + A_i\beta}$$

where,  $A_i$  is given by Equ.(13-77) and  $\beta$  is given by Equ.(13-75).

### Example 13-8

For the current shunt feedback of circuit of Fig.13-24, calculate the current gain without and with feedback.

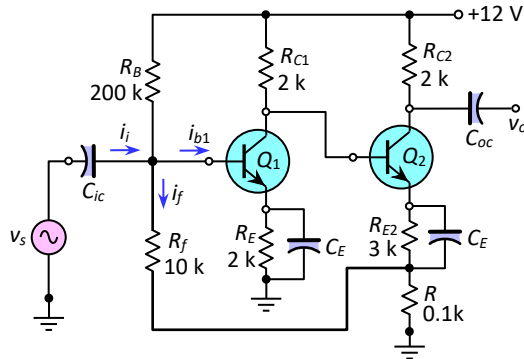


Fig.13-24: Current shunt feedback amplifier

### Solution:

Using Equ.(4.21) 
$$I_{BQ1} = \frac{V_{CC} - V_{BE1}}{R_B + (\beta + 1)R_{E1}} = \frac{12\text{ V} - 0.7\text{ V}}{200\text{ k}\Omega + (100 + 1)2\text{ k}\Omega} \approx 18.77\text{ }\mu\text{A}$$

$$\therefore I_{CQ1} = I_{BQ1} \times h_{fe} = 18.77\text{ }\mu\text{A} \times 100 = 1.88\text{ mA}$$

$$\therefore V_{C1} = V_{CC} - I_{CQ1} \times R_{C1} = 12\text{ V} - 1.88\text{ mA} \times 2\text{ k}\Omega \approx 8.24\text{ V}$$

$$I_{EQ2} = \frac{V_{C1} - V_{BE2}}{R_{E2}} = \frac{8.24\text{ V} - 0.7\text{ V}}{3.1\text{ k}\Omega} \approx 2.43\text{ mA}$$

Therefore, 
$$h_{ie1} \approx h_{fe1} \frac{26\text{ mV}}{I_{CQ1}} = 100 \times \frac{26\text{ mV}}{1.88\text{ mA}} \approx 1.4\text{ k}$$

and, 
$$h_{ie2} \approx h_{fe2} \frac{26 \text{ mV}}{I_{EQ2}} = 100 \times \frac{26 \text{ mV}}{2.43 \text{ mA}} \approx 1 \text{ k}$$

Using Equ.(13-74), 
$$\beta = \frac{R}{R + R_f} = \frac{0.1 \text{ k}}{0.1 \text{ k} + 10 \text{ k}} = 1/101$$

Using Equ.(13-76), 
$$A_i = \frac{h_{fe1} h_{fe2} R_{c1}}{R_{c1} + h_{ie2}} = \frac{100 \times 100 \times 2 \text{ k}}{2 \text{ k} + 1 \text{ k}} = 6666.67 \text{ [Ans.]}$$

Using Equ.(13-77), 
$$A_{if} = \frac{A_i}{1 + A_i \beta} = \frac{6666.67}{1 + 6666.67 \times 1/101} \approx 99.5 \text{ [Ans.]}$$

**Comments:** Due to negative feedback current gain has reduced to 99.5.

## 13.22 Sample Questions

1. What is feedback? Give the classification of feedback with their definitions.
2. Explain the theory of feedback and derive the expressions for the voltage gain of positive and negative feedback.
3. What are positive feedback amplifier and negative feedback amplifiers? Write the advantages and disadvantages of these amplifiers?
4. What is stability? Explain how the negative feedback increases the stability of an amplifier.
5. What is non-linear distortion? Explain how the negative feedback decreases the non-linear distortion in amplifiers.
6. Explain the effect of negative feedback on the lower and higher cut-off frequencies.
7. What do you mean by the bandwidth of an amplifier? Prove that the negative feedback increases the bandwidth of an amplifier.
8. Explain how the negative feedback decreases the noise level in an amplifier.
9. Discuss about the effect of negative feedback on the input and output impedance of an amplifier.
10. Write the names of different feedback topologies and draw their block diagrams.
11. Draw the Voltage series feedback amplifier circuit and derive the expression for the input impedance, output impedance and voltage gain.
12. Draw the voltage shunt feedback amplifier circuit and derive the expression for the voltage gain, input impedance, and output impedance.
13. Prove that the voltage shunt feedback decreases the input impedance of an amplifier. Is it good or bad for the amplifier?
14. Draw the current series feedback amplifier circuit and derive the expression for the input impedance, output impedance and voltage gain.
15. Prove that the current series feedback decreases the gain of an amplifier. Is it good or bad for the amplifier?
16. Prove that the current series feedback increases the output impedance of an amplifier. Is it good or bad for the amplifier- explain why?
17. Draw the current shunt feedback amplifier circuit and derive the expression for the input impedance, output impedance and voltage gain.

18. There are four topologies of feedback configurations in amplifiers. Give a comparison among them.
19. Draw the circuit diagram of a voltage series feedback amplifier using JFET. Calculate its voltage gain, input impedance and output impedance.
20. Draw the circuit diagram of a voltage series feedback amplifier using BJT. Calculate its voltage gain, input impedance and output impedance.
21. Draw a two-stage CE amplifier with negative voltage series feedback. Determine the overall voltage gain of this multi-stage amplifier with feedback.
22. Draw the circuit diagram of a voltage series feedback amplifier using operational amplifier and calculate its voltage gain.
23. Draw a single-stage CE amplifier with negative voltage shunt feedback. With its equivalent circuit, calculate its voltage gain, input impedance and output impedance.
24. Draw the circuit diagram of a voltage shunt feedback amplifier using Op-amp and calculate its voltage gain.
25. Draw the circuit diagram of a current series feedback amplifier using BJT and derive the expression for the voltage gain, input impedance and output impedance.
26. Draw the circuit diagram of a CE amplifier with voltage divider bias and prove that the emitter resistor provide current series feedback.
27. Draw the circuit diagram of a two-stage CE amplifier and apply current shunt feedback. Describe operation procedure of the feedback system.
28. Draw the hybrid-equivalent circuit of the amplifier of question no. 27 and derive the expression for
  - i) The feedback factor
  - ii) Current gain without feedback
  - iii) Current gain with feedback
  - iv) Input impedance with feedback
  - v) Output impedance with feedback

### 13.23 Exercise

1. An amplifier has a voltage gain ( $A$ ) of 96 without feedback. If negative feedback is applied to the amplifier the overall gain ( $A_f$ ) falls to 8. (i) Calculate the feedback factor ( $\beta$ ). (ii) If the value of  $\beta$  remains effective, calculate the required gain of the amplifier ( $A$ ) to make the overall gain  $A_f = 10$ .
2. Negative feedback is applied to an amplifier having open loop gain of 400. The value of feedback factor is 0.2. If the open loop gain of the amplifier decreases by 20%, due to temperature variation, determine the percentage of variation in the closed loop gain.
3. An amplifier has an open-loop mid-band gain of 250. The lower and higher cut-off frequencies of the amplifier are 100 Hz and 500 kHz, respectively. A negative feedback with  $\beta = 0.15$  is applied to the amplifier. Calculate the bandwidth of the amplifier with and without feedback.
4. For the voltage series feedback of Fig.13-25, calculate voltage gain with and without feedback. Assume  $V_P = -3.5$  V, and  $I_{DSS} = 15$  mA for the FET.

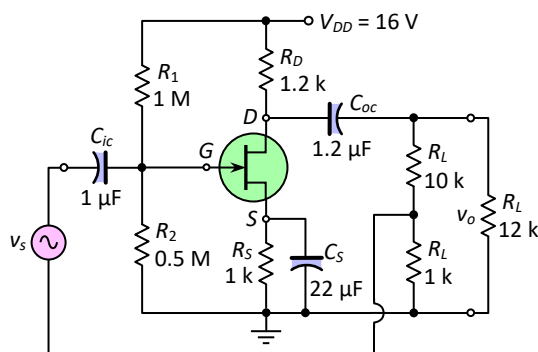


Fig.13-26: Voltage series feedback amplifier for Example 12-4

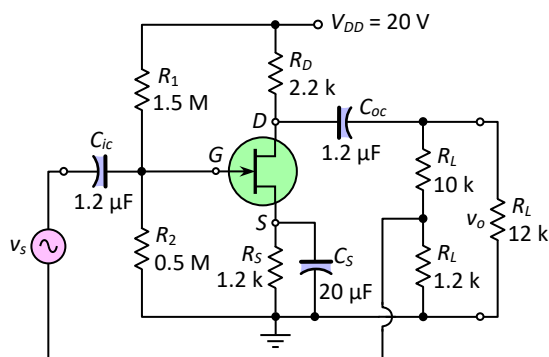


Fig.13-26: Voltage series feedback amplifier for Example 12-4

5. Repeat problem no. 4 for the circuit of Fig.13-26.
6. The direct coupled two-stage CE amplifier of Fig.13-27 utilizes voltage series feedback. Calculate the voltage gain with and without feedback for mid-frequency range. Assume for both transistors  $h_{fe} = 120$ .

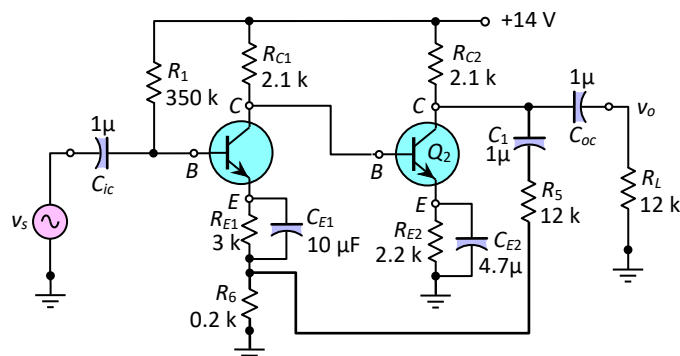


Fig.13-28: Voltage series feedback amplifier for Example 12-5

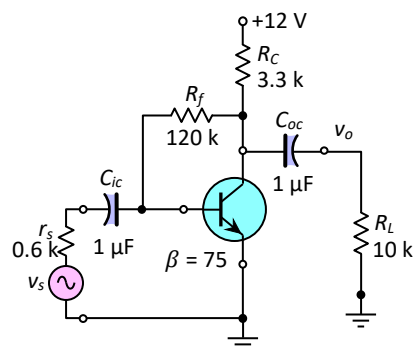


Fig.13-28: Voltage shunt feedback amplifier for Example 12-6

7. For the voltage shunt feedback amplifier of Fig.13-28, calculate overall voltage gain and input impedance. Calculate the lower cut-off frequency, higher cut-off frequency and the bandwidth. Assume
8. For the current series feedback amplifier of Fig.13-29, determine the voltage gain and input impedance without and with feedback in the mid-frequency range.

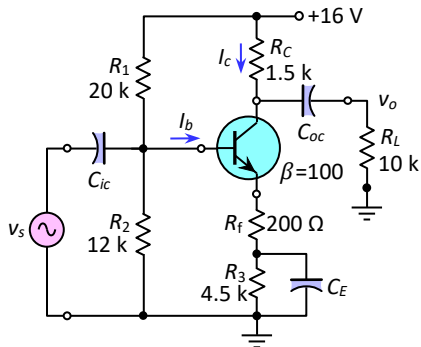


Fig.13-29: Current series feedback amplifier for Example 12-7

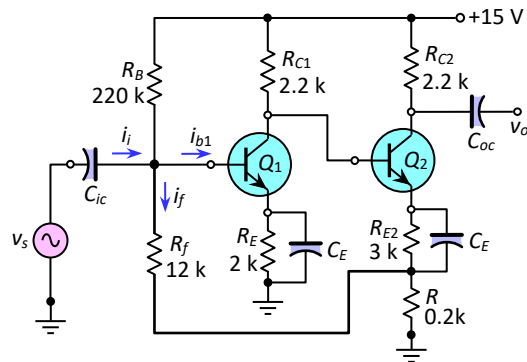


Fig.13-30: Current shunt feedback amplifier

9. For the current shunt feedback of circuit of Fig.13-23, calculate the current gain employing feedback and without employing feedback

====X====.